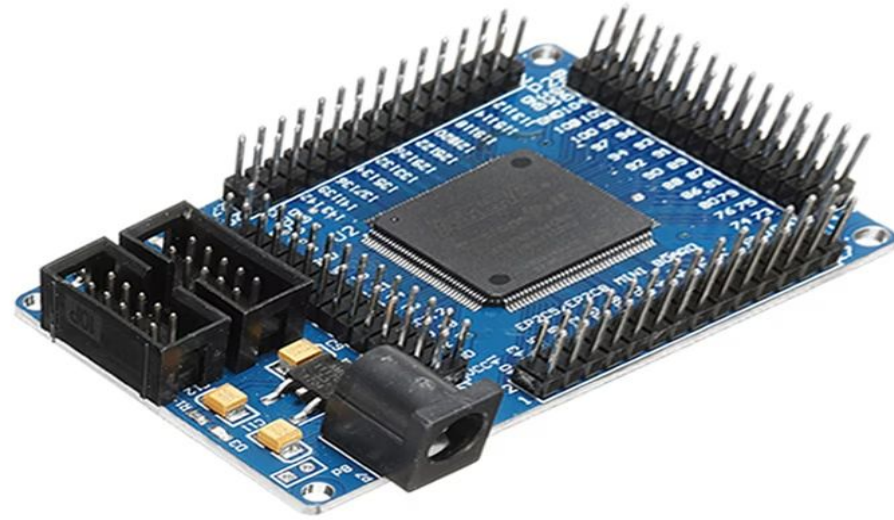


Module 5: FPGA Architecture

Field Programmable Gate Array (FPGA)

- An FPGA is defined as an integrated circuit containing an array of programmable logic blocks, embedded memory, and a matrix of reconfigurable interconnects that allow a designer to implement custom digital hardware after manufacturing.
- Unlike ASICs, FPGAs can be programmed and reprogrammed after manufacturing, making them ideal for prototyping and hardware acceleration.



Key Features

- Reconfigurable hardware
- Parallel processing capability
- High performance
- In-System Programmability (ISP)

FPGA Architecture

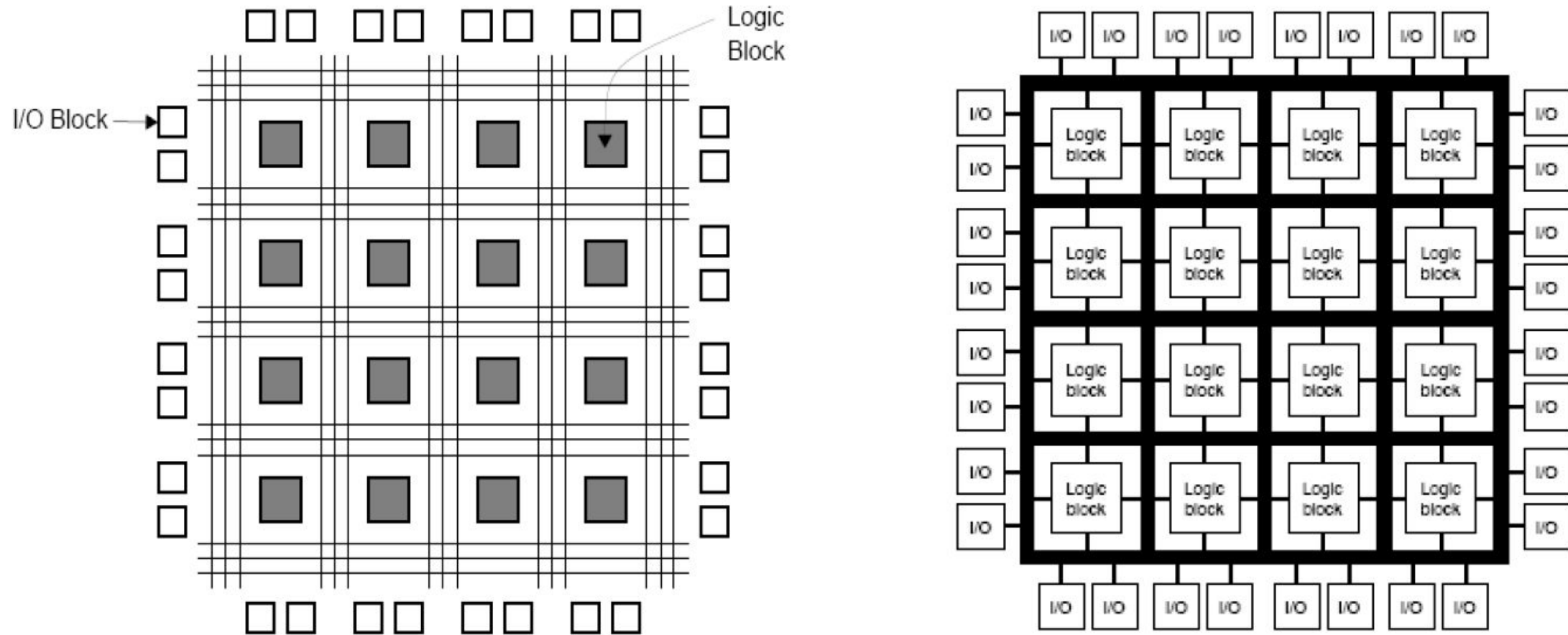
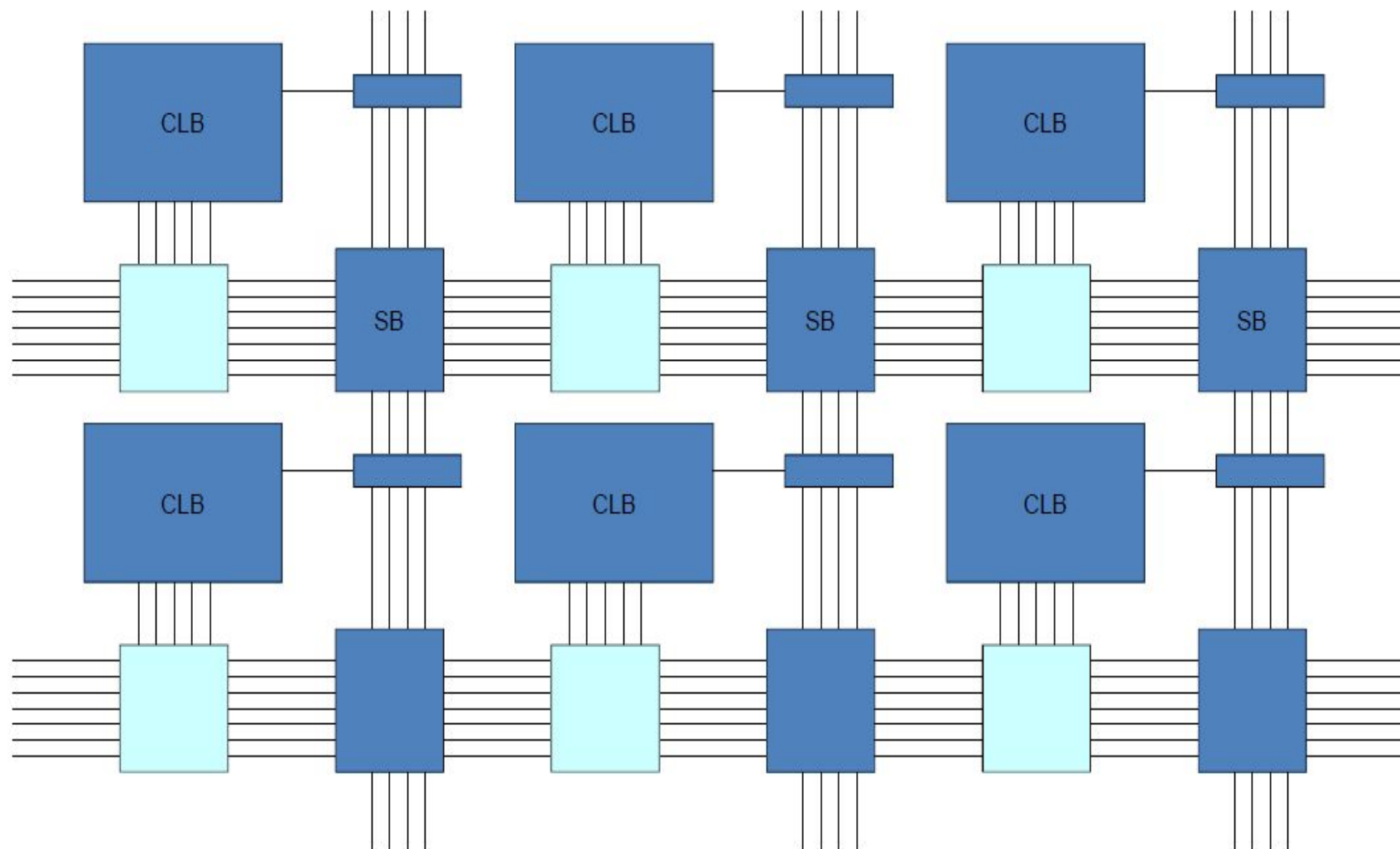


Fig. Structure of an FPGA

FPGA Architecture: More Detailed view



Switch Block Types

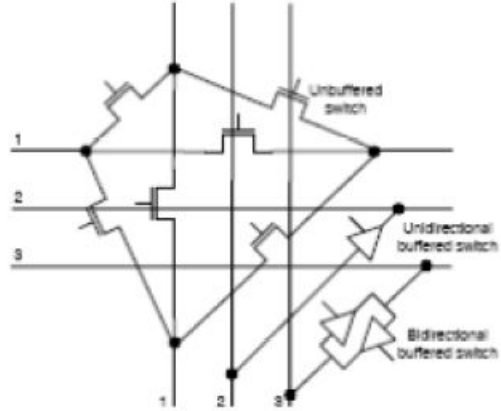


Fig. Switch Block

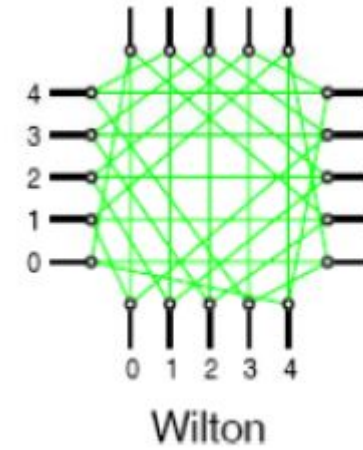
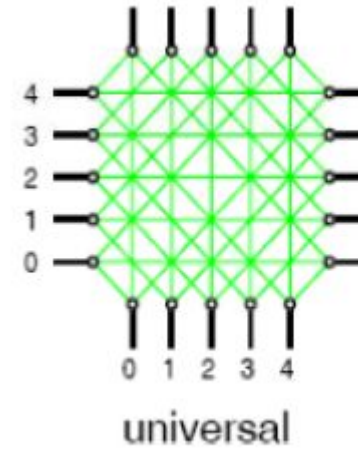
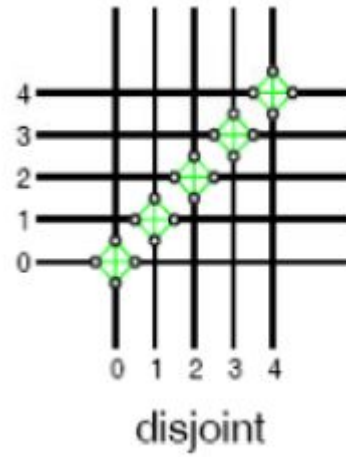
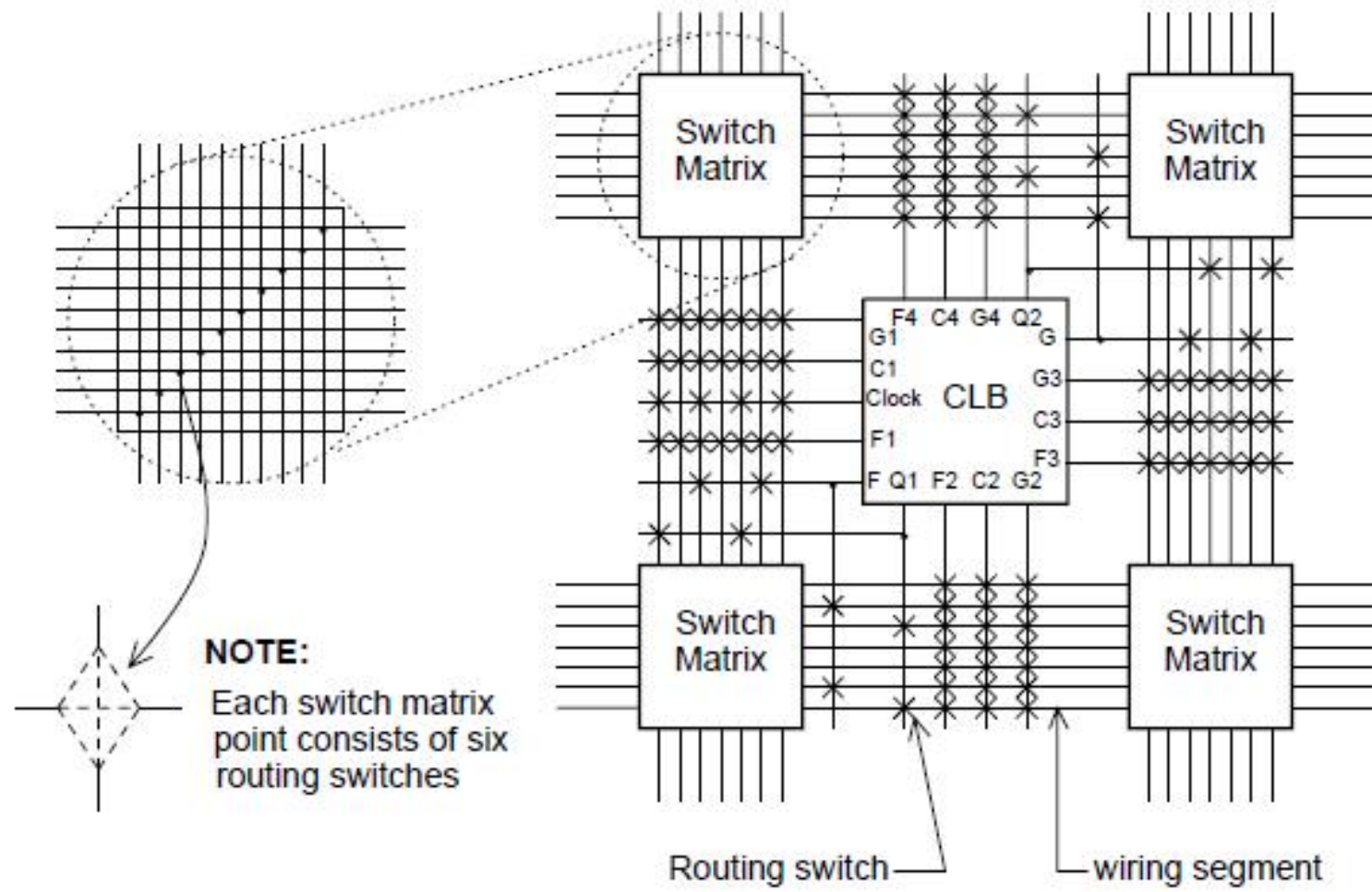


Fig. Types of Switch Block



Types of Switch Block

1. Disjoint Interconnect

- Each input is connected to a limited number of outputs.
- Connections are arranged in a fixed, non-overlapping (disjoint) pattern.
- Requires fewer programmable switches, reducing area and power consumption.
- Offers high speed due to shorter routing paths.
- Least flexible routing architecture.
- Suitable for small and simple digital circuits.

2. Universal Interconnect

- Every input can connect to every output through programmable switches.
- Provides maximum routing flexibility.
- Can implement any connection pattern.
- Requires a large number of switches, increasing chip area and power.
- More complex routing but supports large and complex designs.

3. Wilton Interconnect

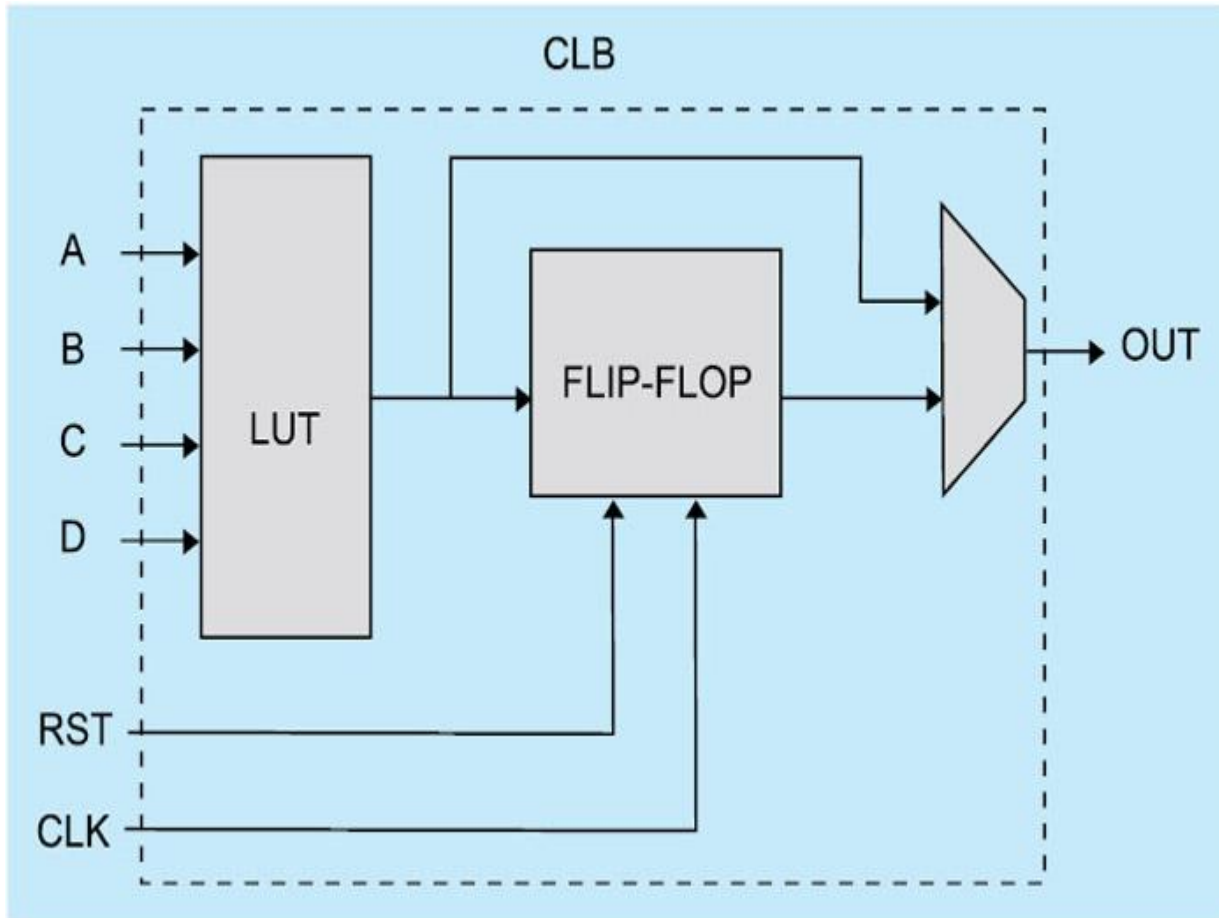
- A modified routing architecture proposed by Steve Wilton.
- Switches are connected in a special permutation pattern instead of full connectivity.
- Provides better routing flexibility than disjoint while using fewer switches than universal.
- Reduces routing congestion and improves channel utilization.
- Achieves better routing success and timing performance.
- Widely used in modern FPGA routing architectures.

Comparison Table

Feature	Disjoint	Universal	Wilton
Routing Flexibility	Low	Very High	High
Number of Switches	Least	Maximum	Moderate
Area Requirement	Low	High	Moderate
Power Consumption	Low	High	Moderate
Performance	Good	Moderate	Best Overall
Applications	Small PLDs	Complex FPGAs	Modern FPGAs

Logic Block Architecture

Configurable Logic Block (CLB)



1. Inputs (A, B, C, D)

- A, B, C, and D are the four logic inputs to the CLB.
- These inputs are provided from the FPGA routing network.
- They are applied to the **Lookup Table (LUT)**.

Purpose:

- Represent variables of the logic function.
- Can implement any Boolean function of up to **4 variables**.

2. Look-Up Table (LUT)

- Look-Up Table (LUT) is fundamentally defined as a programmable memory assembly used to implement Boolean logic functions.

Function

- Implements combinational logic.
- Stores the truth table of the required logic function.
- For a 4-input LUT, there are:
 $2^4 = 16$ possible input combinations.
Therefore, it stores 16 output values.

3. Flip-Flop (FF)

The LUT output is connected to a Flip-Flop.

Function

The flip-flop converts the CLB into a sequential logic element.

It is used for:

- Registers
- Counters
- Shift Registers
- Finite State Machines (FSMs)
- Pipeline Registers

Inputs

CLK (Clock): Determines when data is stored.

RST (Reset): Clears the flip-flop output.

Operation

At every active clock edge:

$Q_{next} = \text{LUT Output}$

Thus, the LUT output is stored in the flip-flop.

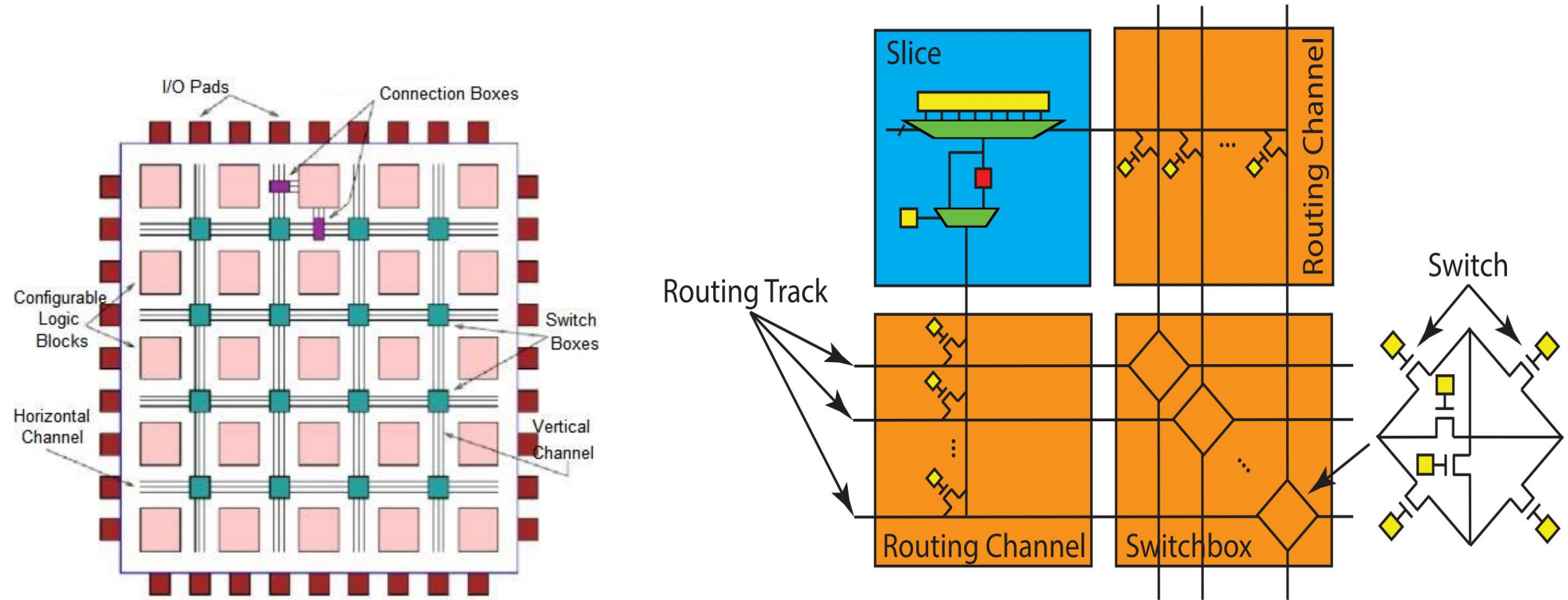
4. Output Multiplexer (MUX)

- The multiplexer selects one of two outputs.

Inputs to the MUX

- Direct LUT output (Combinational path)
- Flip-Flop output (Sequential path)

Routing Architecture

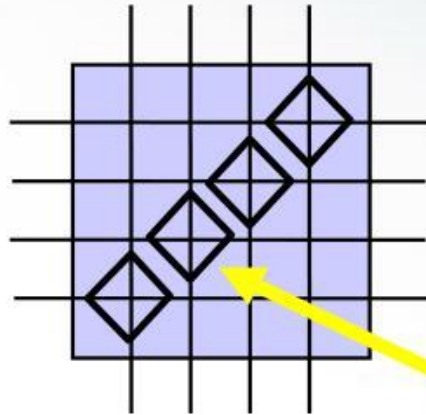


In an FPGA, a **slice** is a small, basic building block of logic.

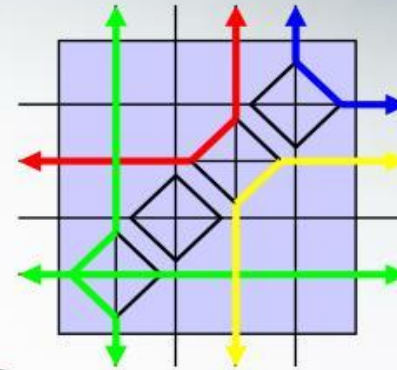
Xilinx (AMD) FPGAs group these slices together inside larger areas called Configurable Logic Blocks (CLBs).

Programmable Interconnections

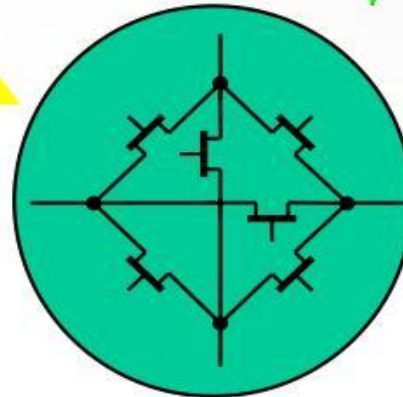
Before Programming



After Programming



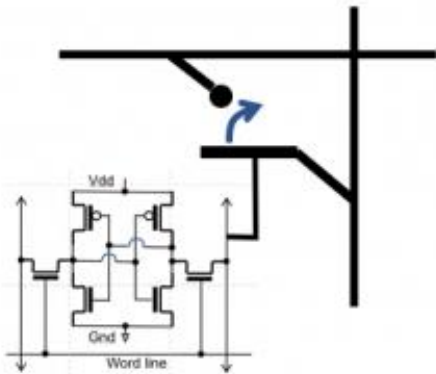
- 6 pass transistors per switch matrix interconnect point
- Pass transistors act as programmable switches
- Pass transistor gates are driven by configuration memory cells



Programmable interconnections

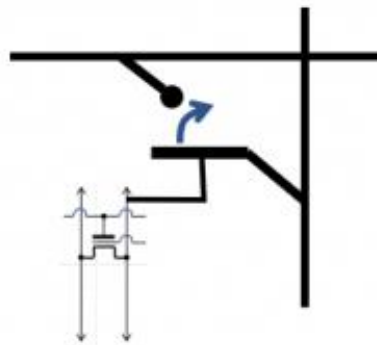
- Programmable interconnections determine how different CLBs communicate.
- FPGA interconnect technologies are SRAM, FLASH, and Antifuse

SRAM



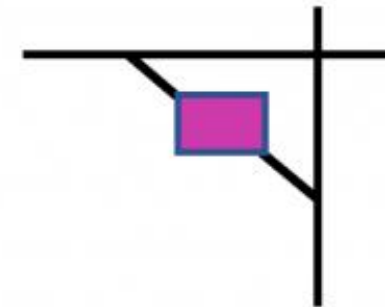
Reprogrammable
and volatile

FLASH



Reprogrammable
and Non-volatile

Antifuse



One-time programmable
and Non-volatile

Programmable interconnections Types

(a) SRAM-Based

- Most common
- Reprogrammable
- Fast configuration
- Used in Xilinx and Altera devices

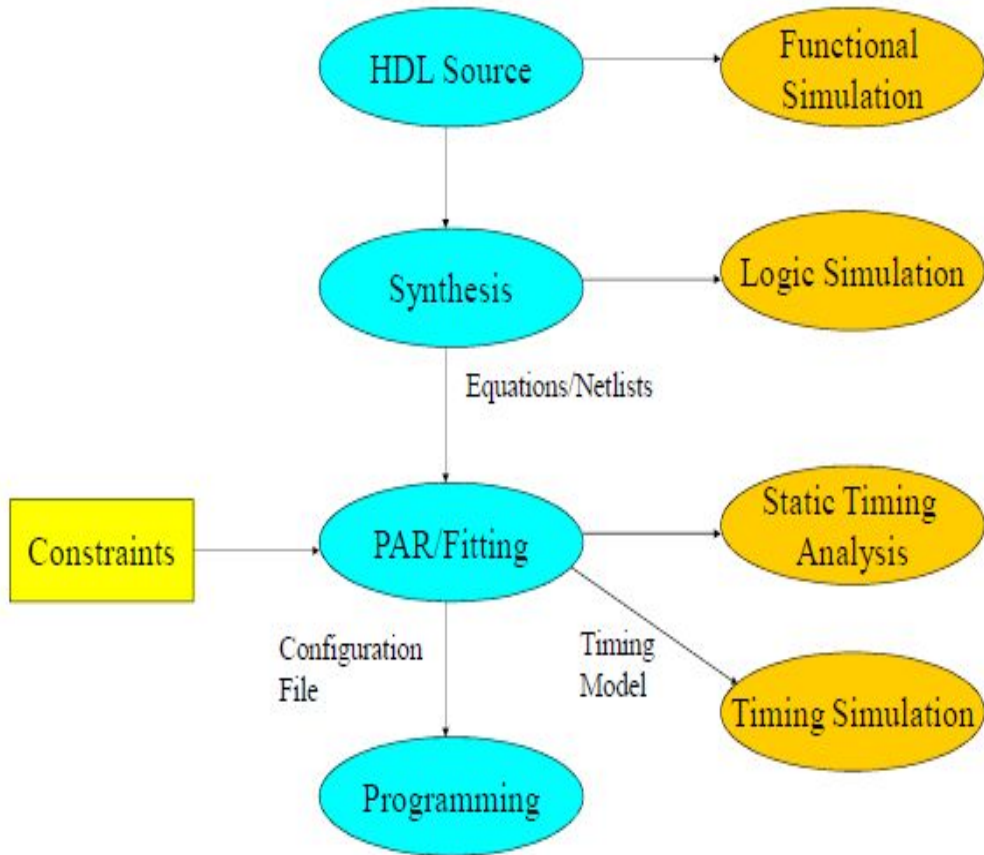
(b) Antifuse-Based

- One-time programmable
- Very low resistance
- High security

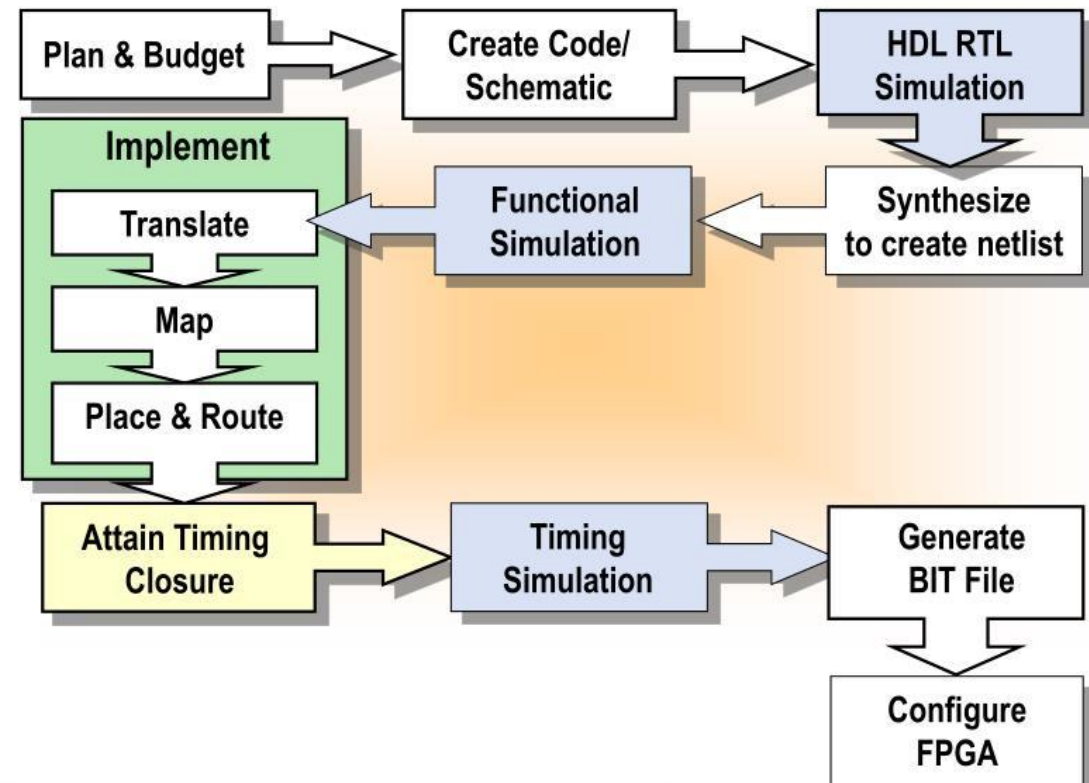
(c) EEPROM/Flash-Based

- Non-volatile
- Retains configuration after power-off

FPGA Design Methodology (Design Flow)



Xilinx Design Flow



HDL Source

The design is written in a Hardware Description Language (HDL) such as **Verilog** or **VHDL**.

It describes the required digital logic, functionality, and behavior of the system.

Functional Simulation

Performed immediately after writing the HDL code.

Verifies the logical functionality of the design without considering hardware delays.

Ensures that the design behaves according to the specifications before synthesis.

Synthesis

The HDL code is translated into a gate-level representation.

The synthesis tool converts the design into **equations/netlists** consisting of logic gates, flip-flops, and other FPGA resources.

This step also optimizes the design for area, speed, and power.

Logic Simulation

Conducted after synthesis using the generated gate-level netlist.

Confirms that the synthesized logic still performs the intended function.

Detects any synthesis-related issues before implementation.

Constraints

The designer specifies implementation requirements such as:

- Clock frequency

- Input/output pin assignments

- Timing constraints

- Area and placement requirements

These constraints guide the implementation tools during Place and Route.

PAR/Fitting (Place and Route)

Placement: Assigns synthesized logic blocks to physical resources (CLBs, LUTs, Flip-Flops) on the FPGA.

Routing: Connects the placed logic blocks using programmable interconnects.

Produces:

- Configuration File (Bitstream):** Used to program the FPGA.

- Timing Model:** Used for timing verification.

Static Timing Analysis (STA)

Analyzes all signal paths without applying input test vectors.

Verifies that setup time, hold time, and clock timing requirements are satisfied.

Ensures that the design can operate at the specified clock frequency.

Timing Simulation

Uses the timing model generated after Place and Route.

Simulates the circuit with actual propagation delays.

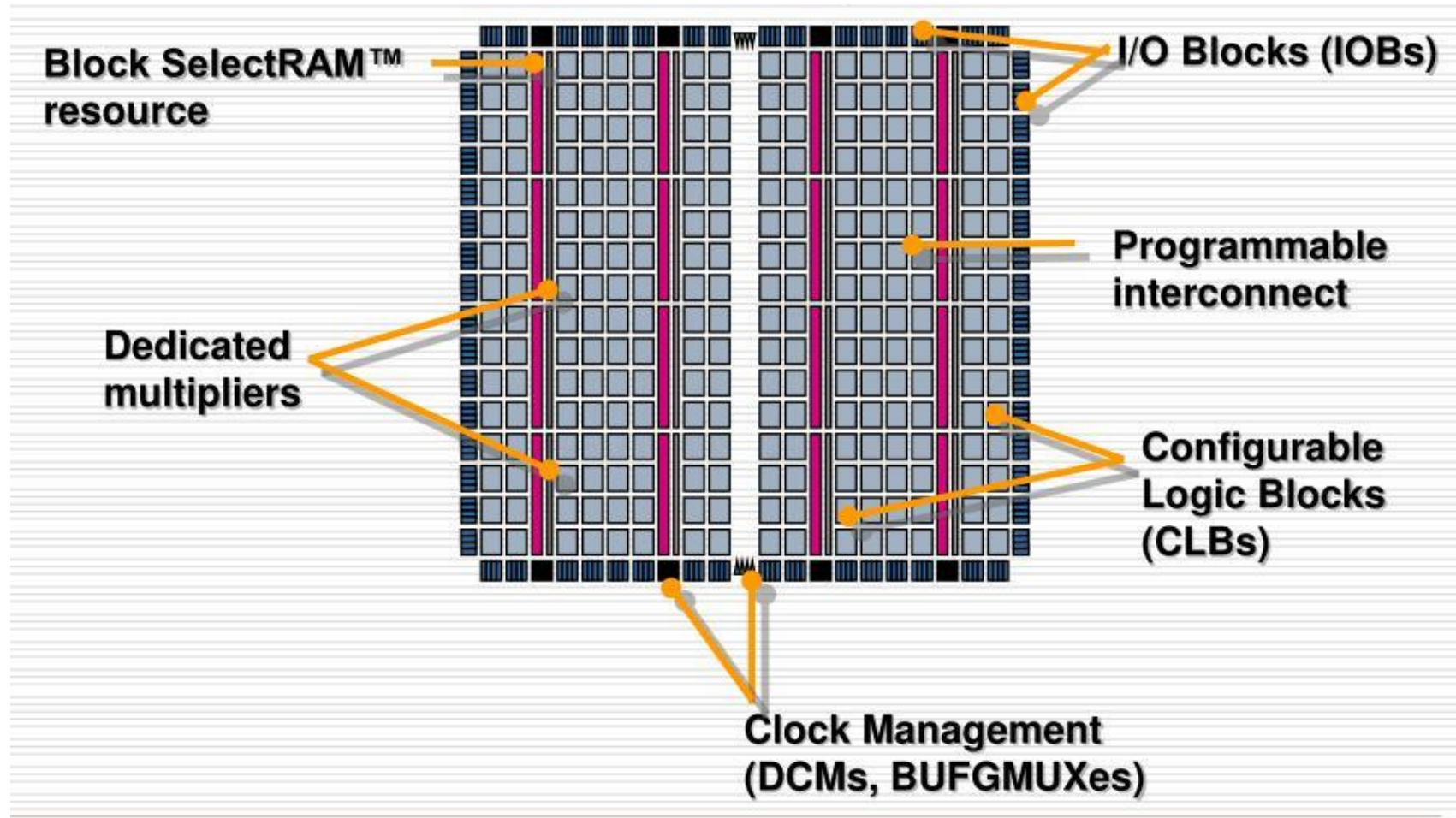
Confirms that the implemented hardware functions correctly under real timing conditions.

Programming

The final configuration file (bitstream) is downloaded into the FPGA.

The FPGA is configured and begins operating according to the implemented design.

Xilinx Virtex-II Architecture



Xilinx Virtex-II Architecture



Xilinx Virtex-II Architecture

- An advanced platform FPGA family built on a 0.15/0.12 μm CMOS process. It features a column-based layout containing Configurable Logic Blocks (CLBs), 18Kb Block RAM (BRAM), 18x18 multiplier blocks for DSP, Digital Clock Managers (DCMs), and programmable Input/Output Blocks (IOBs).
- The Virtex-II family is a high-performance FPGA architecture developed by Xilinx.

Core Architectural Components

- **Configurable Logic Blocks (CLBs):** Contain 4 slices each, featuring 4-input Look-Up Tables (LUTs), fast carry logic, and storage elements/flip-flops to build complex synchronous and combinational circuits.
- **Multiplier Blocks:** Dedicated 18-bit x 18-bit hardware multipliers optimized for high-speed digital signal processing (DSP) without consuming general logic resources.
- **Block RAM:** True dual-port 18Kb memory blocks spread across columns to provide large amounts of on-chip RAM storage.
- **Digital Clock Managers (DCMs):** Provide self-calibration, frequency synthesis, and phase shifting for robust clock management.
- **Input/Output Blocks (IOBs):** Support high-speed single-ended and differential signalling standards like LVDS, LVPECL, and bus-oriented DDR memory interfaces.

Abbreviation	Full Form	Description
LVDS	Low Voltage Differential Signalling	A high-speed, low-power differential signalling standard that transmits data using a small voltage difference between two wires. It offers high noise immunity and is widely used in FPGA interfaces, LCD displays, cameras, and communication systems.
LVPECL	Low Voltage Positive Emitter-Coupled Logic	A high-speed differential signalling standard derived from ECL technology. It provides very low jitter, high switching speed, and is commonly used in clock distribution, high-speed networking, and FPGA clock inputs.
DDR	Double data Rate	A memory interface technology that transfers data on both the rising and falling edges of the clock signal, effectively doubling the data transfer rate compared to SDR (Single Data Rate).

XILINX VIRTEX-II FPGA ARCHITECTURE

CONFIGURABLE LOGIC BLOCK (CLB)

- Contains Logic Cells (Slices)
- Look-Up Tables (LUTs)
- Flip-Flops
- Carry Logic
- Wide Function Multiplexers

BLOCK RAM

- 18 Kbits true dual-port Block RAM
- Used for data storage and FIFOs
- Efficient memory resources

DCM (DIGITAL CLOCK MANAGER)

- Frequency Synthesis
- Phase Alignment
- Clock Skew Correction
- Multiple Clock Outputs

SWITCH MATRIX

- Programmable Routing
- Connects CLBs, Block RAMs, IOBs and DCMs
- Built using Pass Transistors and Buffers

ARCHITECTURAL LEGEND

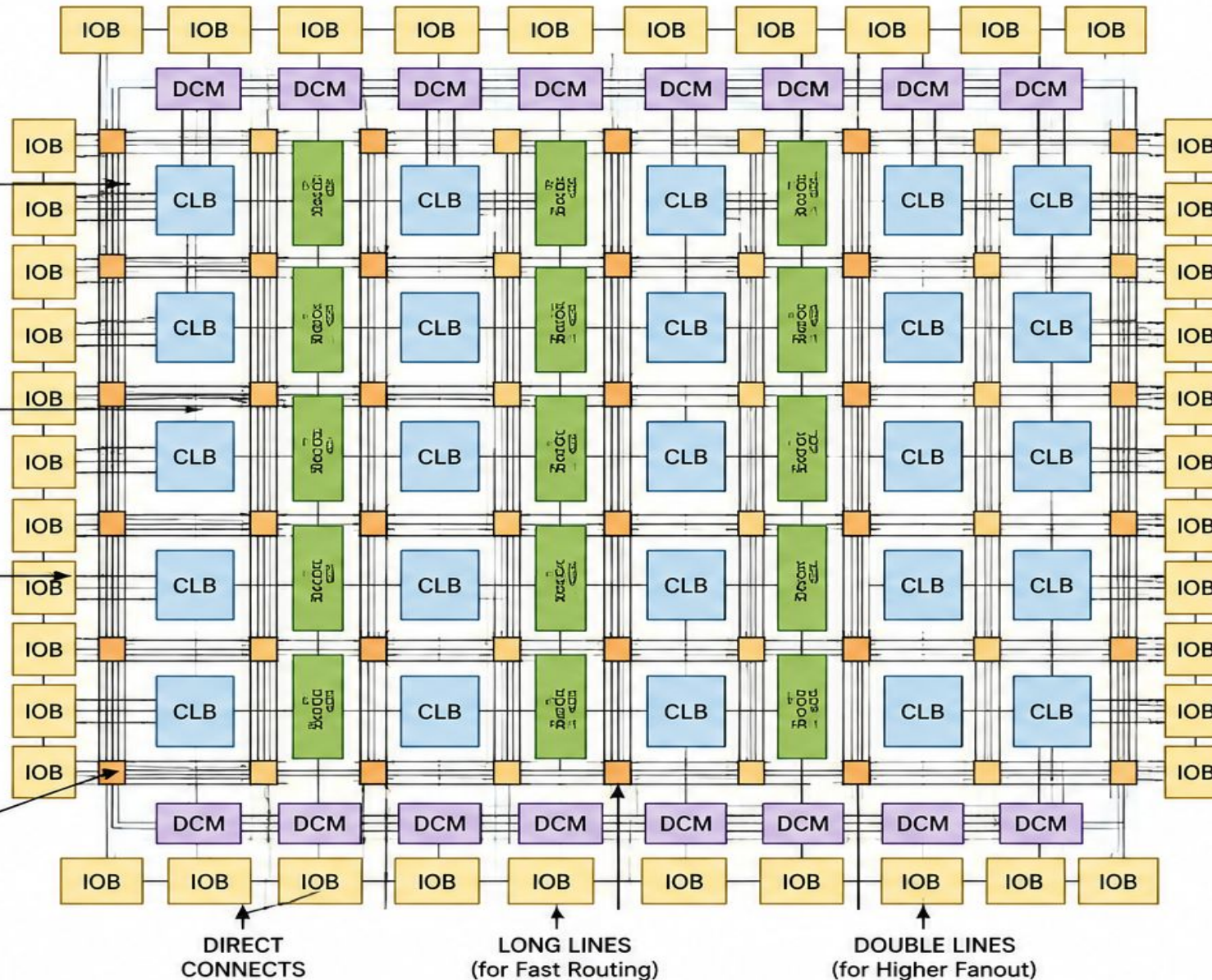
- CLB (Configurable Logic Block)
- Block RAM
- DCM (Digital Clock Manager)
- IOB (Input/Output Block)
- Switch Matrix / Routing

IOB (INPUT/OUTPUT BLOCK)

- Interfaces FPGA with external pins
- Supports multiple I/O standards
- Includes registers for input/output operations
- Programmable impedance

HIGHLIGHTS OF VIRTEX-II ARCHITECTURE

- High Performance (up to ~300 MHz)
- 0.15 μm CMOS Technology
- Dedicated Block RAM
- Flexible and Hierarchical Routing
- Optimized for Speed and Density
- Supports Advanced Multipliers (DSP Blocks) in Larger Devices



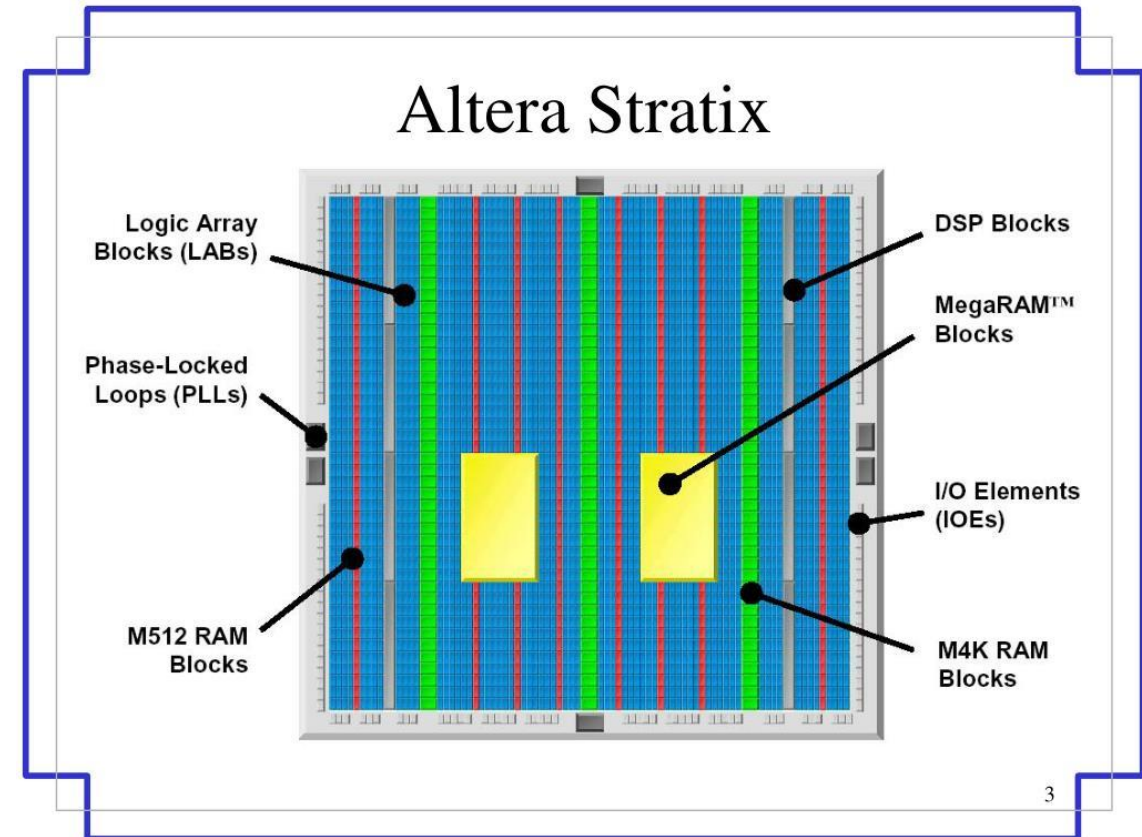
Virtex-II FPGA architecture is based on a symmetric, island-style design with a 2D array of Configurable Logic Blocks (CLBs) surrounded by I/O Blocks (IOBs) and interconnected through a programmable routing network.

Altera Stratix Architecture

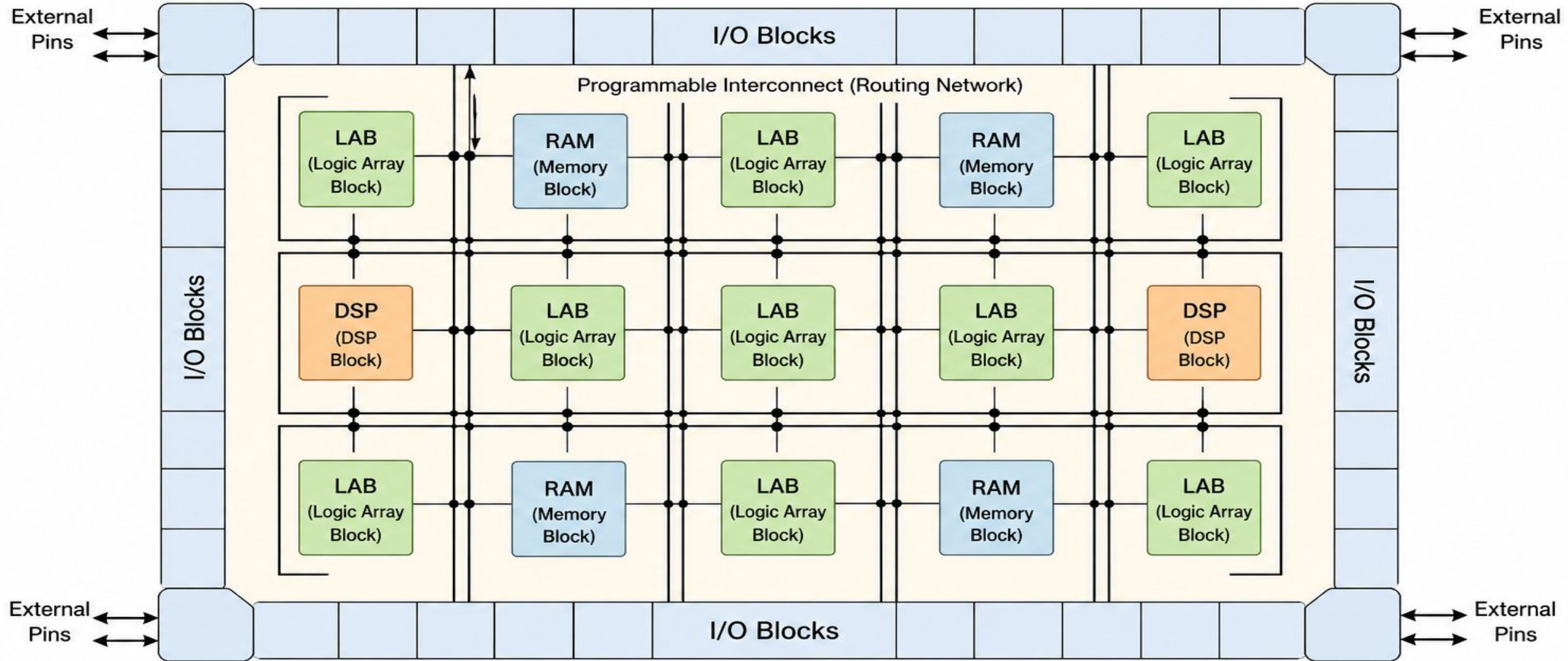
Altera Stratix is a family of high-performance FPGA (Field-Programmable Gate Array) devices developed by Altera (now Intel FPGA). Stratix devices are designed for high-speed digital systems and provide programmable logic, memory, DSP blocks, high-speed interconnects, and I/O resources.

The major components are:

1. Logic Array Blocks (LABs)
2. Adaptive/Logic Elements
3. Programmable Interconnection Network
4. Embedded Memory Blocks
5. DSP Blocks
6. I/O Elements
7. Clock Management Resources
8. High-Speed Transceivers in applicable Stratix families



ALTERA STRATIX FPGA – BASIC ARCHITECTURE



LAB Logic Array Block
(Contains Logic Elements,
LUTs, Registers)

RAM Embedded Memory
Block (RAM/ROM/
FIFO, etc.)

DSP DSP Block
(Multipliers, Adders,
Accumulators)

I/O I/O Blocks
(Programmable I/O
Resources)

—●— Programmable
Routing
Network